Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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1. TEAM AND PROJECT OVERVIEW

Team Name

Taygood

Project / Solution Name

Kutip: cash-collection assistant for the Orbyte hub

Name (Leader)

Hansen Yap Hong San

Email & Phone (Leader)

yaphongsan@gmail.com · Phone: **TODO**

University Affiliation(s)

Taylor's University

Live Prototype URL

Not yet deployed; runs locally from the repo (Vercel planned). **TODO: add URL**

Code Repository URL (public / access granted to judges)

https://github.com/YapHongSanHansen/Xmum (public)

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The user opens Kutip in a browser: a Next.js 16 / React 19 app shown as a module inside a simulated Orbyte hub, with Owner and Clerk roles. Synthetic data (invoices, outflows, bank-statement lines, customer messages and slip images) loads from a seed into a Zustand store kept in the browser.

Every state change goes through a pure TypeScript rules engine: cash forecast and early warning, credit check, chase-plan ranking, the exact-quote check, bank-line matching and approval rights. Each step is written to an append-only activity log.

When a customer reply arrives, the browser calls the app's server route /api/extract (local today; Vercel is the planned host). The route sends the message text and slip image to the Anthropic API (Claude Opus 5.5) and validates the JSON reply with Zod. On any error or timeout it returns a cached extraction labelled "cached result". The fields go back to the engine, which decides; the AI never changes money state.

Approved reminders open WhatsApp through wa.me click-to-chat links, so a person presses send. The preliminary build uses no database (a hosted one is planned for Phase 2). The API key exists only as a server environment variable.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9	Language for the app, rules engine and tests	Apache-2.0
Next.js 16.4 and React 19.3	Web framework, UI and the server route for the AI call	MIT
Tailwind CSS 4.3 and lucide-react 1.61	Styling and icons	MIT; ISC
Zustand 5 and Zod 4.6	Client state store; validation of the AI's JSON output	MIT
@anthropic-ai/sdk 1.3	Client for the Anthropic API	Apache-2.0
Vitest 5.0 and ESLint 9	Unit tests (97 passing) and linting	MIT

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Anthropic Messages API (Anthropic)	Bukti: reads a customer reply and slip image into structured fields with Claude Opus 5.5	Paid, pay-as-you-go. Optional: cached results are used without a key
WhatsApp click-to-chat links, wa.me (Meta)	Opens WhatsApp with an approved reminder prefilled; a person presses send. A link, not an API call	Free
No other external APIs	Bank data, accounting data and Orbyte Connect are simulated with synthetic CSV data	Not applicable

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel (planned)	Will host the Next.js app and the /api/extract route, with the API key as an environment variable. Not yet deployed	Plan and region not yet selected
GitHub	Public code repository and commit history	Free
Anthropic API (cloud inference)	Runs Claude Opus 5.5 for Bukti	Pay-as-you-go; hosted by Anthropic outside Malaysia
Orbyte (Alpha Red Solutions)	Sponsor workspace: a synthetic Kutip client and project record; no runtime integration	Sponsor account; region not stated
No database or authentication service	Demo state lives in the browser and resets from the seed; roles are simulated	Not applicable

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Claude Opus 5.5 (claude-opus-5-5)	Anthropic, via API	Pre-trained, used as-is with a prompt and JSON schema; no fine-tuning	Anthropic Commercial Terms of Service

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Kutip demo seed (Lori Laju Logistics)	This repo: web/src/engine/seed.ts	Synthetic, team-created	Apache-2.0 (this repo)	N
Synthetic payment slip (fictional "Bank Cempaka")	This repo: web/public/slips/	Synthetic, team-created	Apache-2.0 (this repo)	N
Cached AI extractions for the 3 demo messages	This repo: web/src/engine/cached.ts	Synthetic, team-created	Apache-2.0 (this repo)	N
Experian State of Credit 2025 (figures cited)	experian.com.my, 21 May 2025	Public report; cited in docs only, not loaded in the app	Experian copyright	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude Code (Anthropic)	Research, PRD, code, tests, docs, submission drafts	Major
OpenAI Codex desktop	Repository review; drafting the declaration	Minor

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Team members reviewed and adapted AI-generated code and documents before committing them. The rules engine is covered by 97 Vitest tests that replay every seed check and the full demo script, including the expected activity log. Research figures were spot-checked against their sources, with corrections marked in place. Inside the product, AI output is never trusted directly: each field must quote the customer's exact words, low-confidence items go to a person, and payments need a matching bank line.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Alpha Red Solutions	Orbyte briefing	Shaped our simulated hub frame
Alpha Red Solutions	Orbyte platform access (no API, credits or data provided)	Explored modules and roles; created a synthetic Kutip client and project record. The prototype calls no Orbyte API

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Not essential. The prototype runs fully without Orbyte: the hub frame, roles, approvals and Connect page are simulated inside our app, as the brief allows. Orbyte access only informed the design so that Kutip fits the hub. In a real deployment Kutip would use Orbyte Connect for accounting and bank data, Orbyte roles for permissions and the Orbyte audit trail; our planned invoice-source adapter is where that integration would go.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No real user data is collected. The prototype uses synthetic data only: every business, person, invoice, bank line, message and slip is fictional, and slips use a made-up bank. Demo state is kept in the viewer's browser and cleared by Reset demo; we run no database. When Bukti reads a message, the synthetic text and slip image are sent to the Anthropic API over HTTPS; the key stays on the server.

For a real deployment under Malaysia's PDPA 2010 (as amended in 2024): the SME is the data user and Kutip/Orbyte its processor; contact data is used only to follow up invoices; clerks never see cash figures; extraction keeps only needed fields and masks account numbers; slip images are encrypted with an owner-set retention limit; the AI provider must not train on customer data; cross-border processing is disclosed, with in-region hosting preferred.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

- **Misreading:** the model may misread an amount or invoice number, or invent a quote. Mitigation: each field must quote the message exactly or it is dropped; fields under 0.85 confidence go to a person; nothing is marked paid without a matching bank line; AI never writes customer messages.
- **Ambiguous Manglish, BM and Chinese** (for example "next Friday"): handed to a person with one question instead of guessed.
- **Single provider, outages, rate limits, latency:** a 25-second timeout and cached extractions keep the demo working; the AI step sits behind one server route, so the provider can be swapped.
- **Unmeasured accuracy:** a 30-50 message test set with a 95% exact-amount target is planned; the live AI path is less tested than the cached path.
- **Data residency:** the provider is hosted outside Malaysia.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [ ] The prototype described here is accessible to judges through the links provided in Section 1.
- [ ] All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

[Date] 25 October 2026